

IOT 2025/2026 - Challenge 3

Part 2: exercise

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Data provided)

$f_{carrier} = 868 \text{ MHz}$ (carrier frequency)

$BW = 125 \text{ kHz}$ (bandwidth)

$N = 40$ (number of sensors)

$P = 20 \text{ B} = 160 \text{ b}$ (message payload)

$\lambda = 2 \text{ packet/min}$ (intensity of transmission – Poisson – process)

EQ1)

I have modelled the transmissions as an ALOHA based network following the request of simplification made by the text.

The total arrival rate to the gateway is: $L = N \cdot \lambda = 80 \text{ packet/min} = 1.33 \text{ packet/s}$.

We know that the probability of success is $P_{success} = e^{-2 \cdot L \cdot T}$, where T is the packet airtime so, if we force $P_{success} \geq 0.75$, we get the following condition: $e^{-2 \cdot L \cdot T} \geq 0.75$.

So the condition on the packet airtime that must be satisfied is: $T \leq \frac{\ln(0.75)}{-2 \cdot L} = 108 \text{ ms}$.

Using the suggested online calculator (<https://www.thethingsnetwork.org/airtime-calculator>) I have determined that only SF7 allows us to have a packet airtime smaller than T ($71.9 \text{ ms} \leq 108 \text{ ms}$) while all the other spreading factors generate airtimes bigger than T .

Answer: SF7.

EQ2)

The reason of the difference between the performance of the nodes is that some nodes are physically closer to the gateway while some others are more distant. In case of collisions between packets, the nearest transmitter always wins because its signal is stronger (has less attenuation with respect to the one sent by a distant node). So the only valuable solution is to move the nodes closer to the gateway (to improve the overall performance) and also be sure that the distance from the gateway is nearly uniform across all of them.

Answer: 2 – Move the nodes closer to the gateway.